DNA Damages, DNA Repair Mechanism and Introduction to Mutations

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Objectives:

At the end of this chapter, it is expected to;

- Describe the various types of DNA damages
- Elaborate and appreciate the various mechanism of DNA repair
- Explain the impact of defect in DNA repair mechanisms
- Differentiate various kinds of Mutations

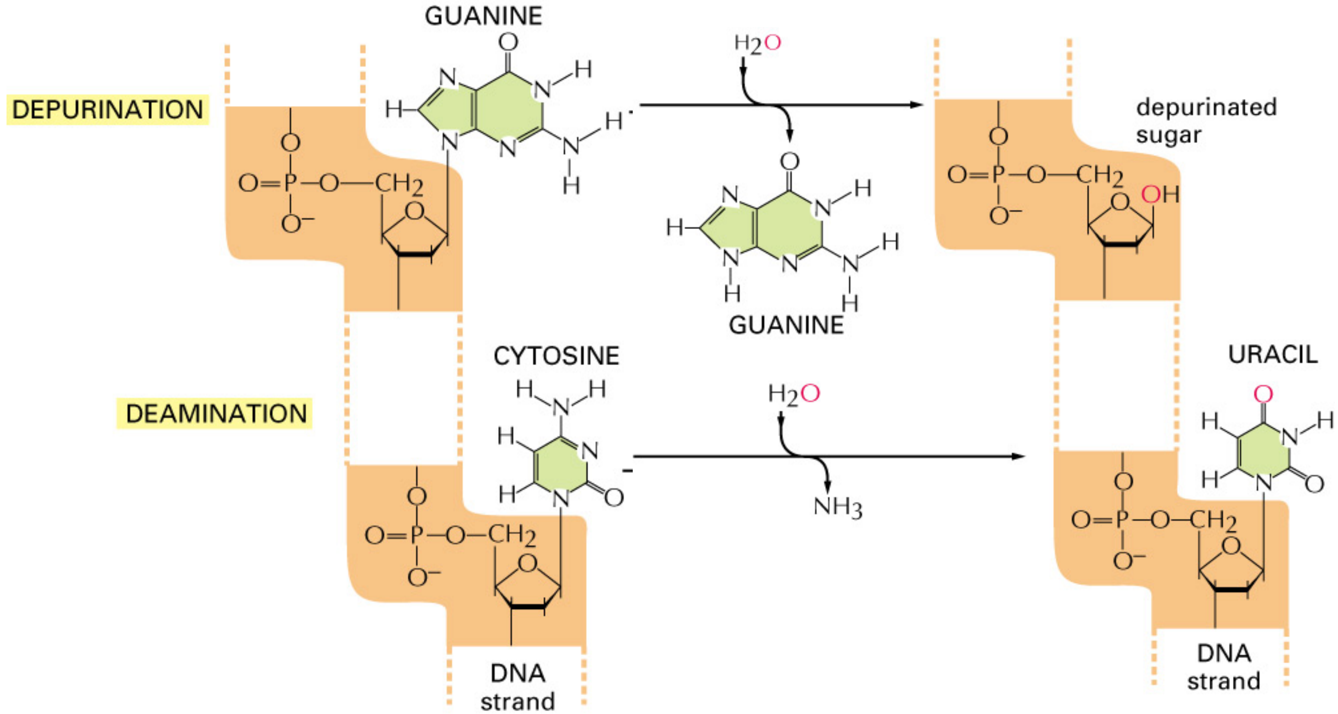
Types of DNA Damage

1. Deamination
2. Depurination
3. Dimerization
4. Alkylation
5. Oxidative damage
6. Replication errors
7. Double-strand breaks (DSB)

Types of DNA Damages

I. Single-base alteration
   A. Depurination
   B. Deamination of cytosine to uracil
   C. Deamination of adenine to hypoxanthine
   D. Alkylation of base
   E. Insertion or deletion of nucleotide
   F. Base-analog incorporation
II. Two-base alteration
   A. UV light–induced thymine-thymine (pyrimidine) dimer
   B. Bifunctional alkylating agent cross-linkage
III. Chain breaks
   A. Ionizing radiation
   B. Radioactive disintegration of backbone element
   C. Oxidative free radical formation
IV. Cross-linkage
   A. Between bases in same or opposite strands
   B. Between DNA and protein molecules (eg, histones)

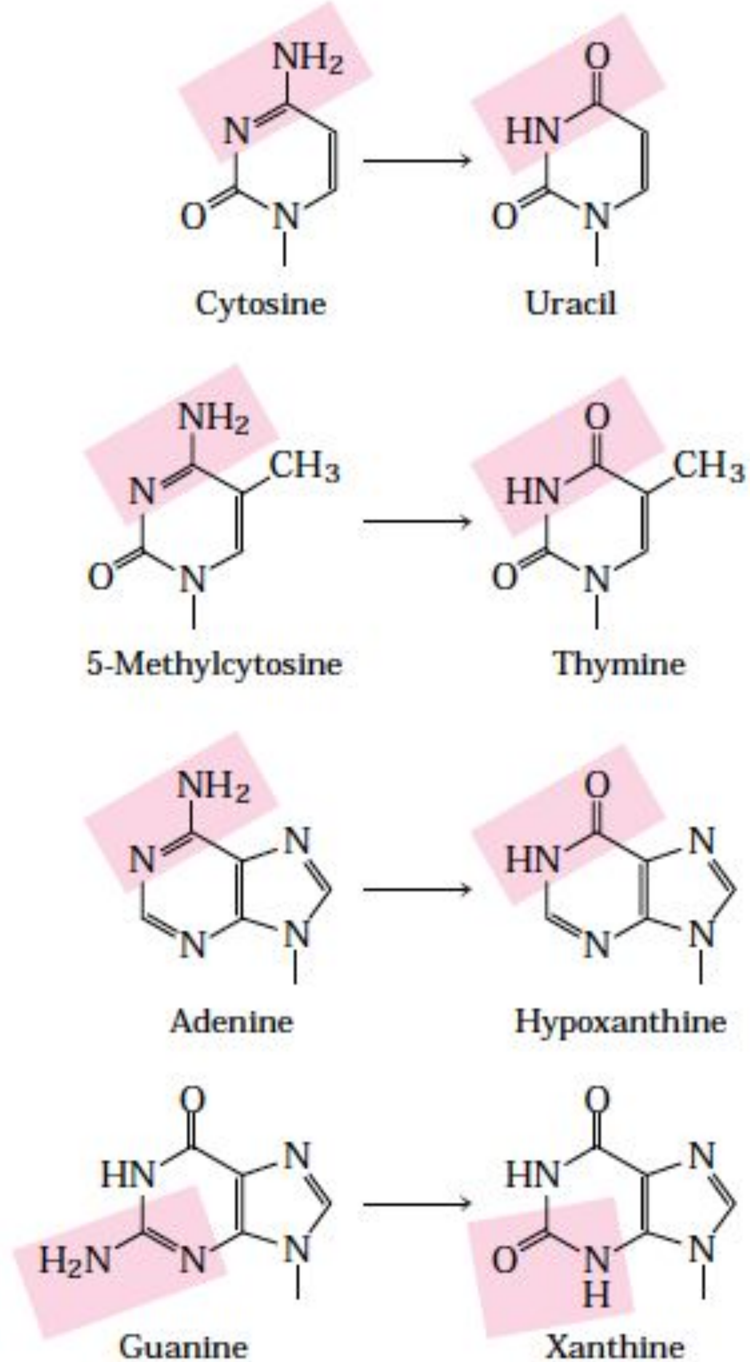
Common types of DNA damage -- 1



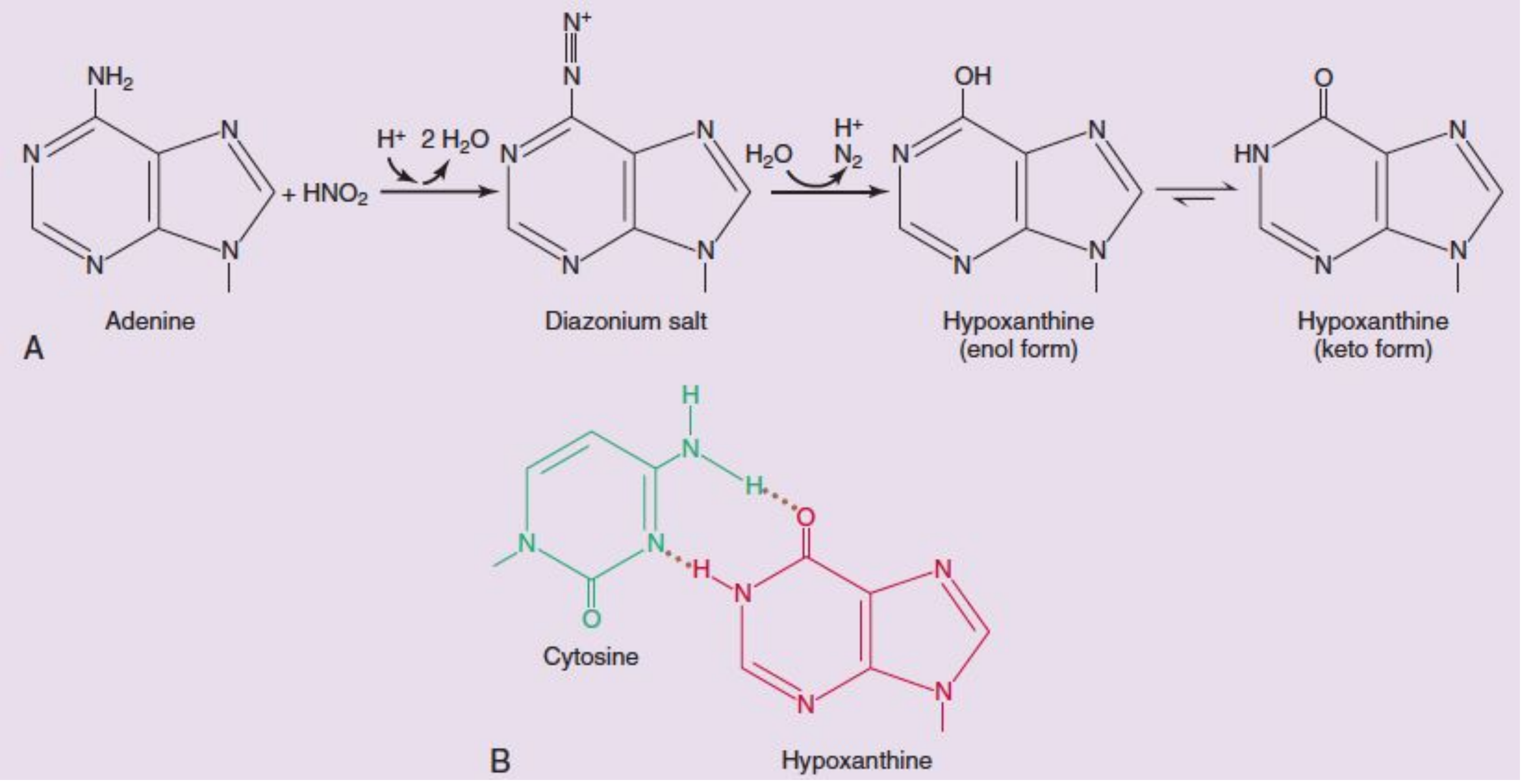
1. **Depurination: Loss of A, G**
   - Hydrolysis of the N-glycosyl bond between the base and the pentose.
   - Occurs at a higher rate for purines than for pyrimidines.
   - One in 10^5 purines are lost from DNA every 24 hours.

2. Deamination: C --> U, A --> Hypoxanthine

- Several nucleotide bases undergo spontaneous loss of their exocyclic amino groups (deamination).
- Deamination of cytosine (in DNA) to uracil occurs in about one of every 10^7 cytidine residues in 24 hours.
- Deamination of adenine and guanine occurs at about $1/100^{th}$.
- Precursors of nitrous acid (nitrosamines, nitrite and nitrate salts), which promotes deamination reactions.



Action of a deaminating agent



A - Reaction of HNO_2 with adenine.
B - Hypoxanthine pairs with cytosine instead of thymine.

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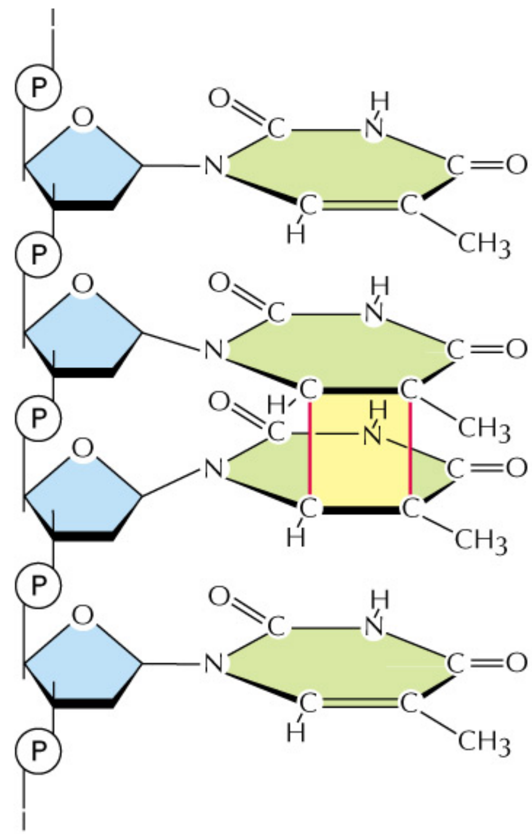
Common types of DNA damage -- 2



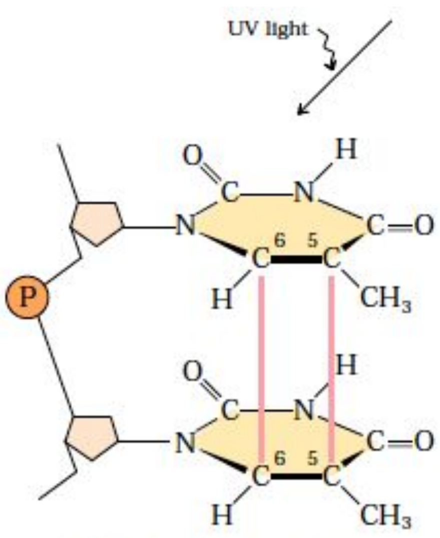
Pyrimidine dimers (UV induced).

Pyrimidine dimers: (UV/wavelengths of 200 to 400 nm induced)

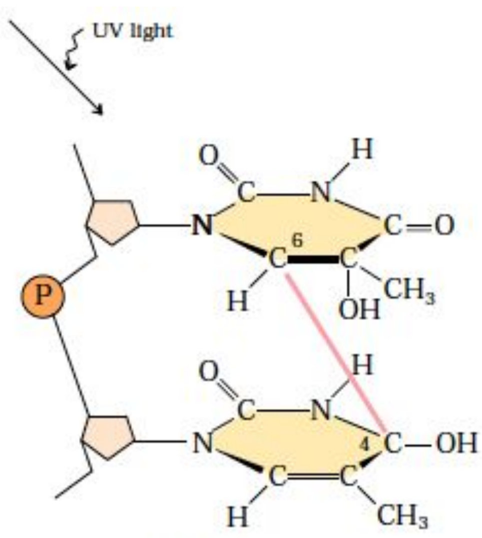
(i) Cyclobutane pyrimidine dimer: happens most frequently between adjacent thymidine residues on the same DNA strand.

(ii) 6-4 photoproduct (a second type pyrimidine dimer) is also formed during UV irradiation.

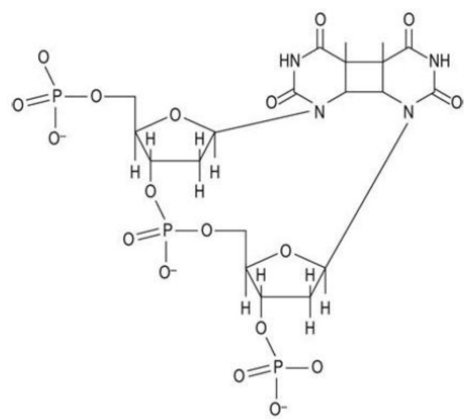
(iii) UV and ionizing radiations are responsible for about 10% of all DNA damage caused by environmental agents.

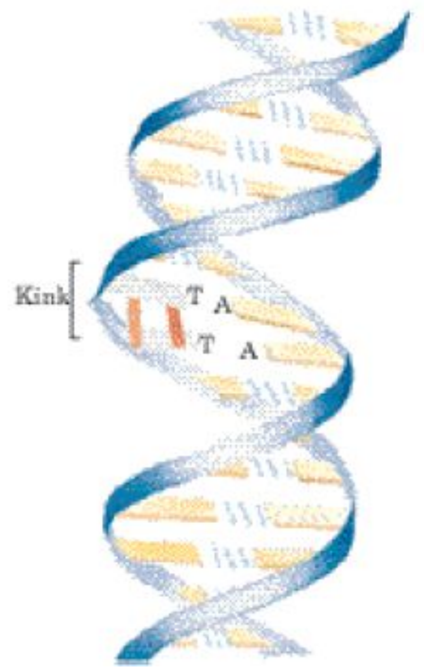


Cyclobutane thymine dimer



6-4 Photoproduct





Common types of DNA damage -- 3

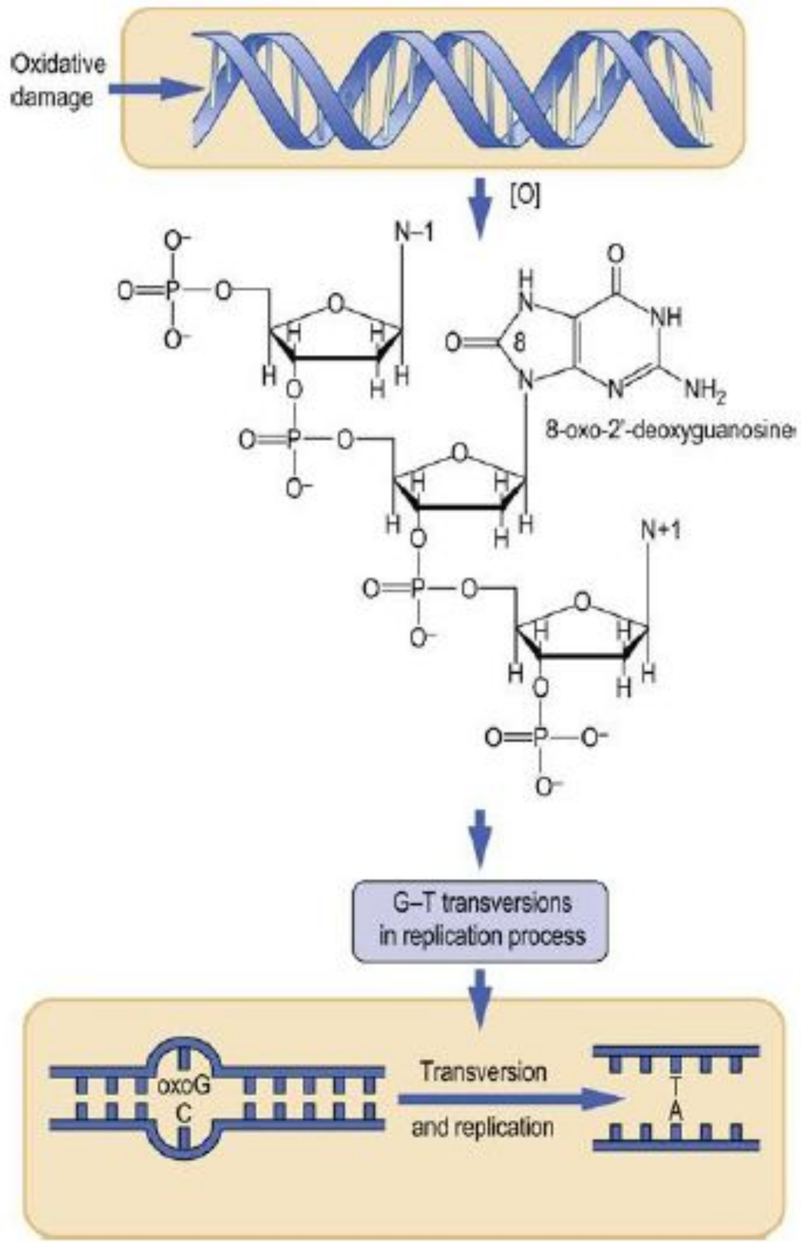
Alkylation:

- Alkylating agents – dimethylsulfate, ethylnitrosourea, nitrosamines, mustard gas and vinyl chloride.
  - An alkyl group (e.g., CH_3) gets added to bases; chemical induced; some harmless, some cause mutations by mispairing during replication or stop polymerase altogether

Oxidative damage of DNA:

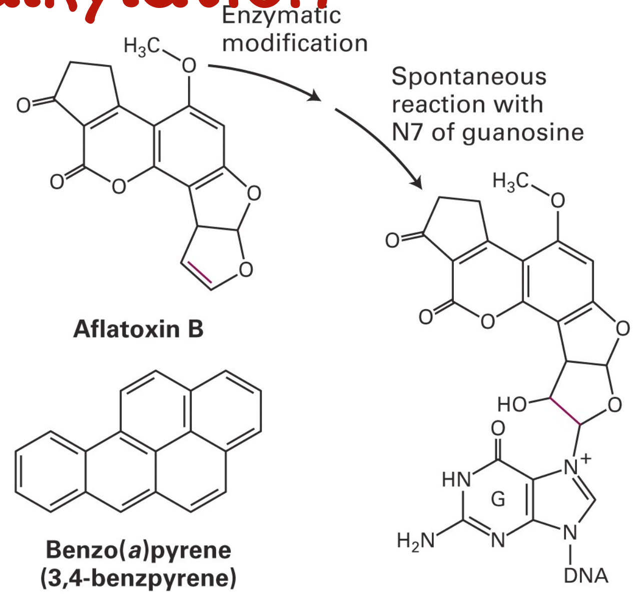
- Excited-oxygen species such as hydrogen peroxide, hydroxyl radicals, and superoxide radicals arise during irradiation or as a byproduct of aerobic metabolism, are responsible for oxidative DNA damage.

Oxidative damage to DNA



Common types of DNA damage -- 3

Two carcinogens that mutate (the P53 gene) by base alkylation



+ Mismatches (mistakes in DNA synthesis)
Interstrand cross-links,
Double-strand DNA breaks

Total damage from all mechanisms: 10^4 - 10^6 lesions/day!

DNA Repair

A fundamental difference from RNA, protein, lipid, etc.

All these others can be replaced, but DNA must be preserved

Cells require a means for repair of missing, altered or incorrect bases, bulges due to insertion or deletion, UV-induced pyrimidine dimers, strand breaks or cross-links

Diverse DNA repair systems

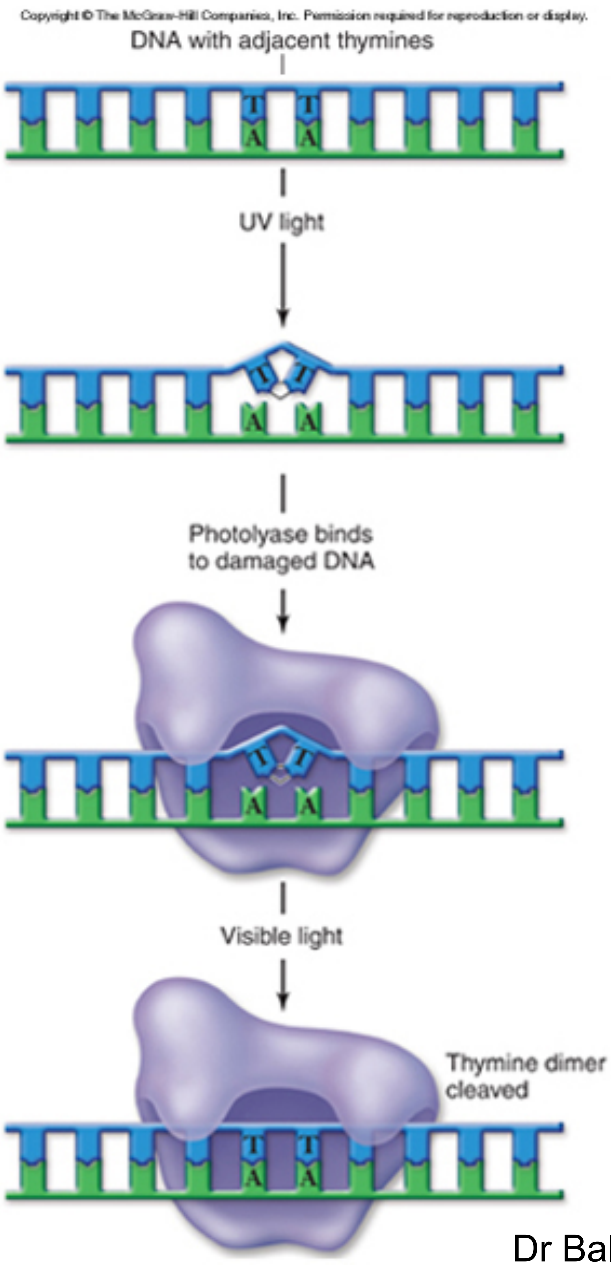
General mechanisms:

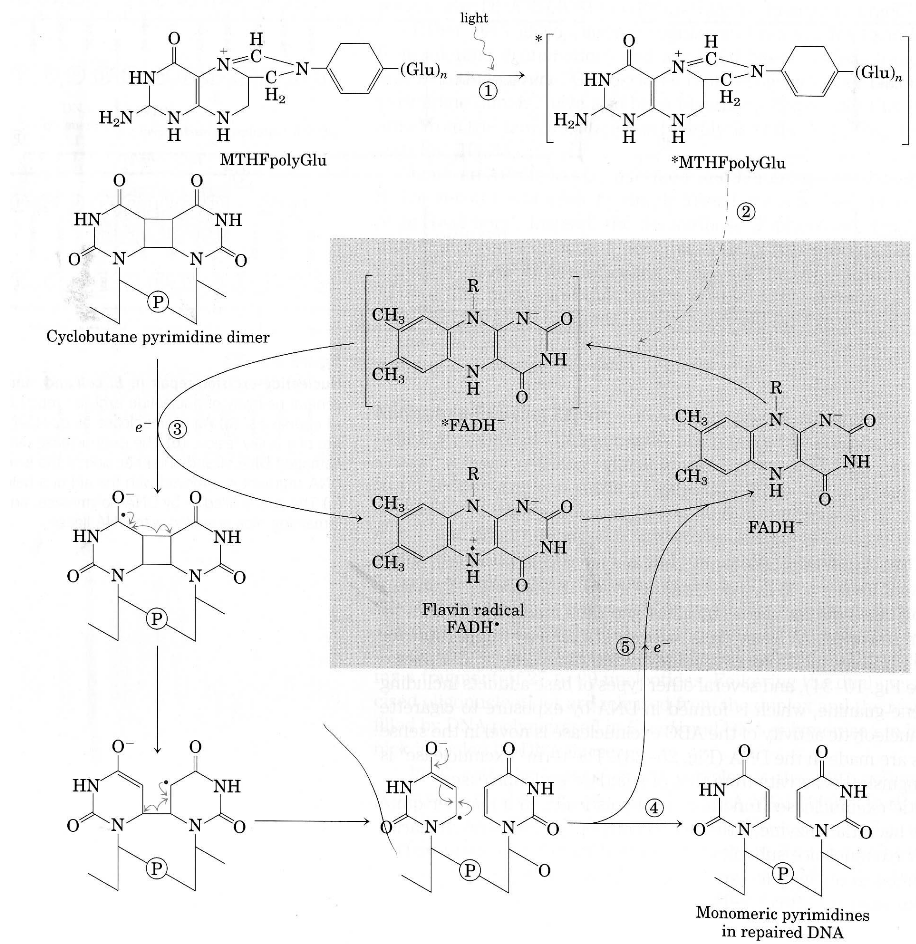
1. Direct repair, e.g. pyrimidine dimers
2. Excision repairs
   a. Base excision repair
   b. Nucleotide excision repair
3. Mismatch excision repair
4. Double-strand break repair and recombination

Direct repair (Photo reactivation)

- Also called light induced repair.
- The photo reactivating (PR) enzyme brings about an enzymatic cleavage of thymine dimers activated by the visible light (300-600 nm) leading to a restoration of the monomeric condition.
- Enzyme is found in all organisms.
- Binds to “NN” dimers, including cytosine dimers, in the DNA.
- Activated by visible light energy and brings about cleavage of “C-C” bonds of the dimer.

Direct repair (Photo reactivation)





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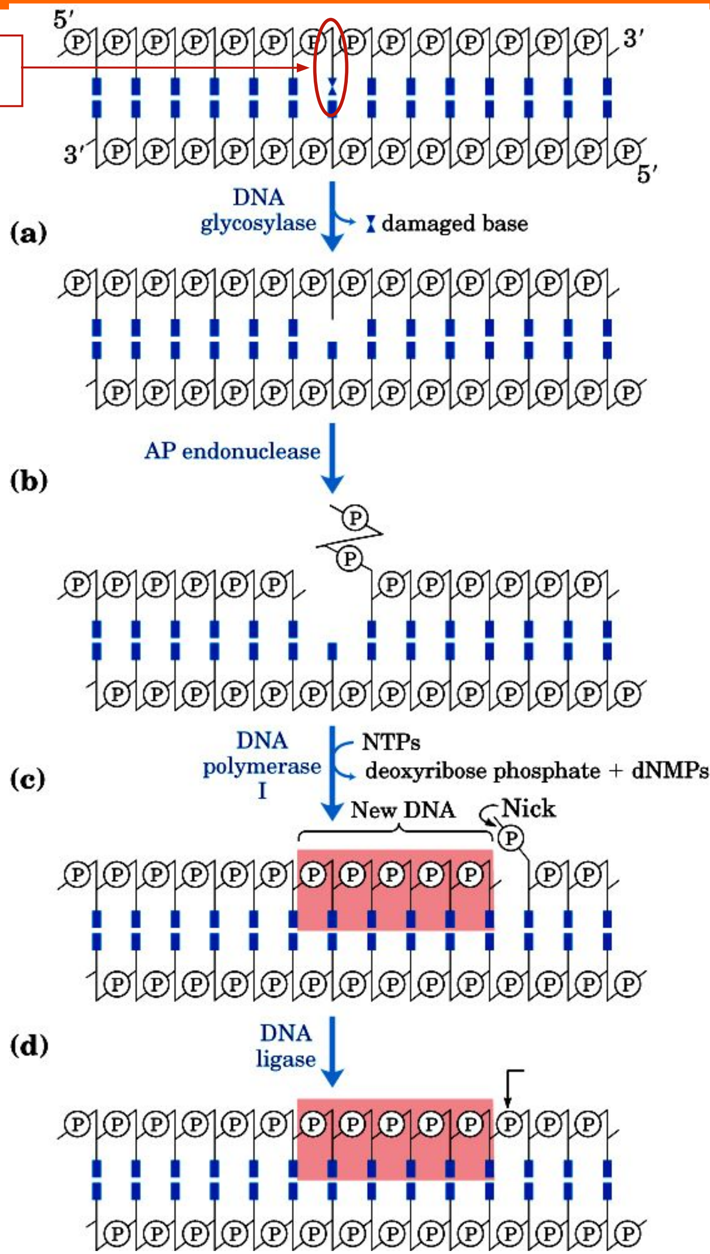
Excision repairs

- **Base excision repair**
- **Nucleotide excision repair**

Base excision repair

Damaged base



Base excision repair pathway (BER).

(a) A DNA glycosylase recognizes a damaged base and cleaves between the base and deoxyribose in the backbone.

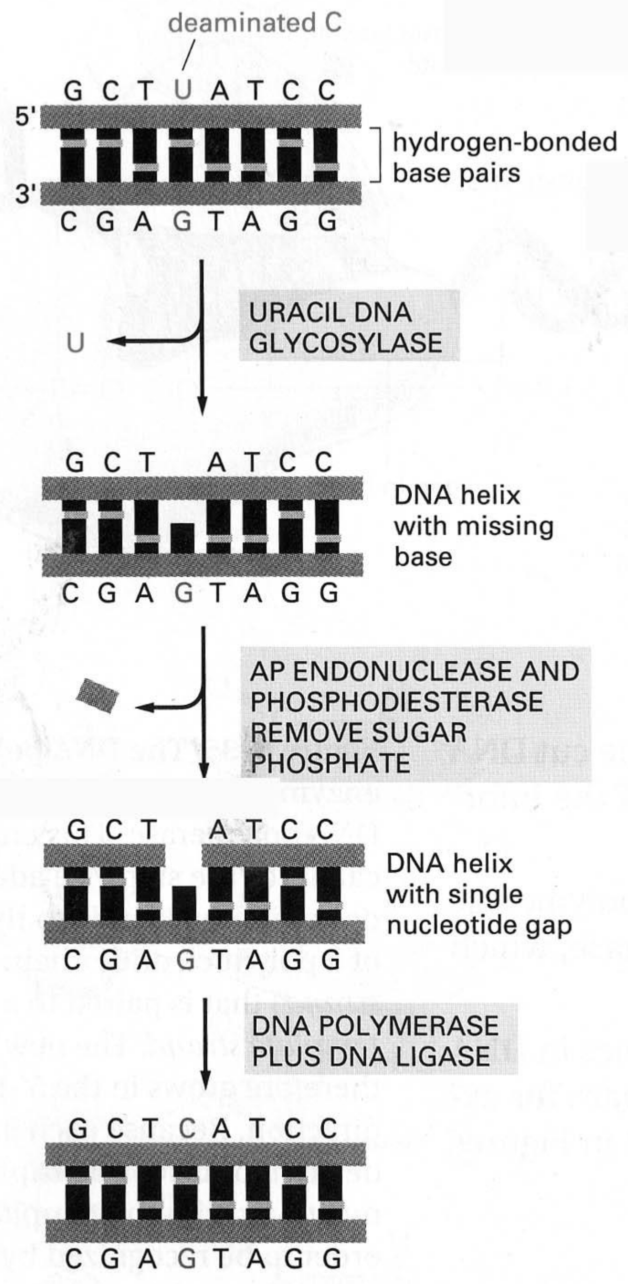
(b) An AP endonuclease cleaves the phosphodiester backbone near the AP site.

(c) DNA polymerase I initiates repair synthesis from the free 3′ OH at the nick, removing a portion of the damaged strand (with its 5′ → 3′ exonuclease activity) and replacing it with undamaged DNA.

(d) The nick remaining after DNA polymerase I has dissociated is sealed by DNA ligase.

AP= apurinic or apyrimidinic (a=without)

Base Excision repair



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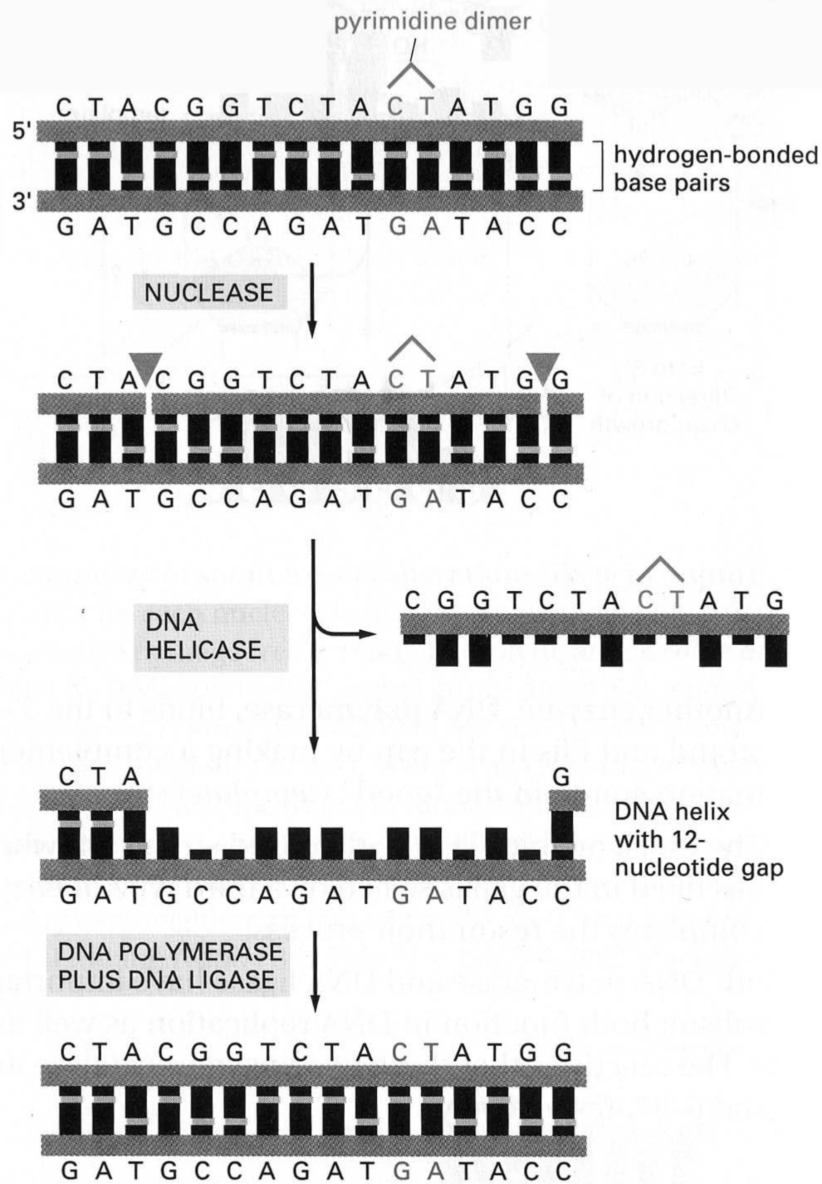
Nucleotide excision repair



Nucleotide excision repair pathway:

(a) A multienzyme complex recognize the the bulky lesions such as pyrimidine dimers.

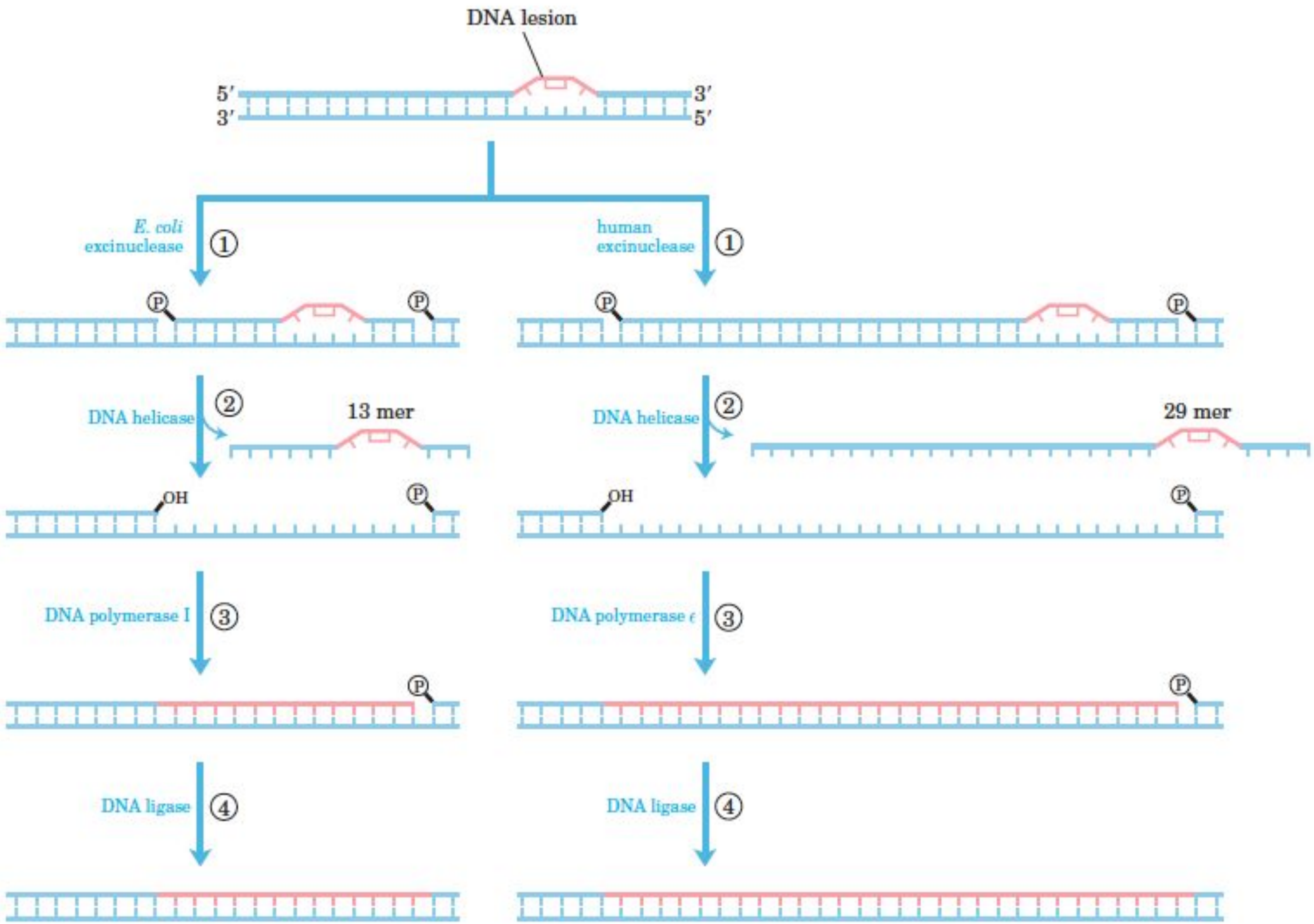
(b) One cut is made on each side of the lesion, and an associated helicase enzyme then removes entire damaged portions.

(c) The multienzyme complex in bacteria leaves the gap of 12bp and in human it is more than twice in size.

(d) The gap then filled by DNA polymerase from the free 3' OH at the nick.

(e) The nick remaining after DNA polymerase has dissociated is sealed by DNA ligase.

Nucleotide excision repair



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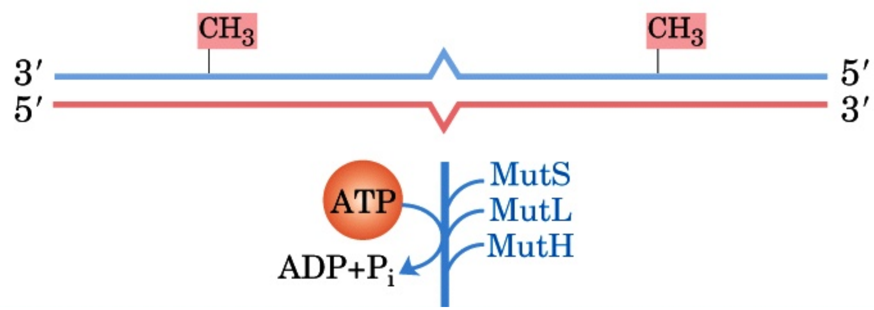
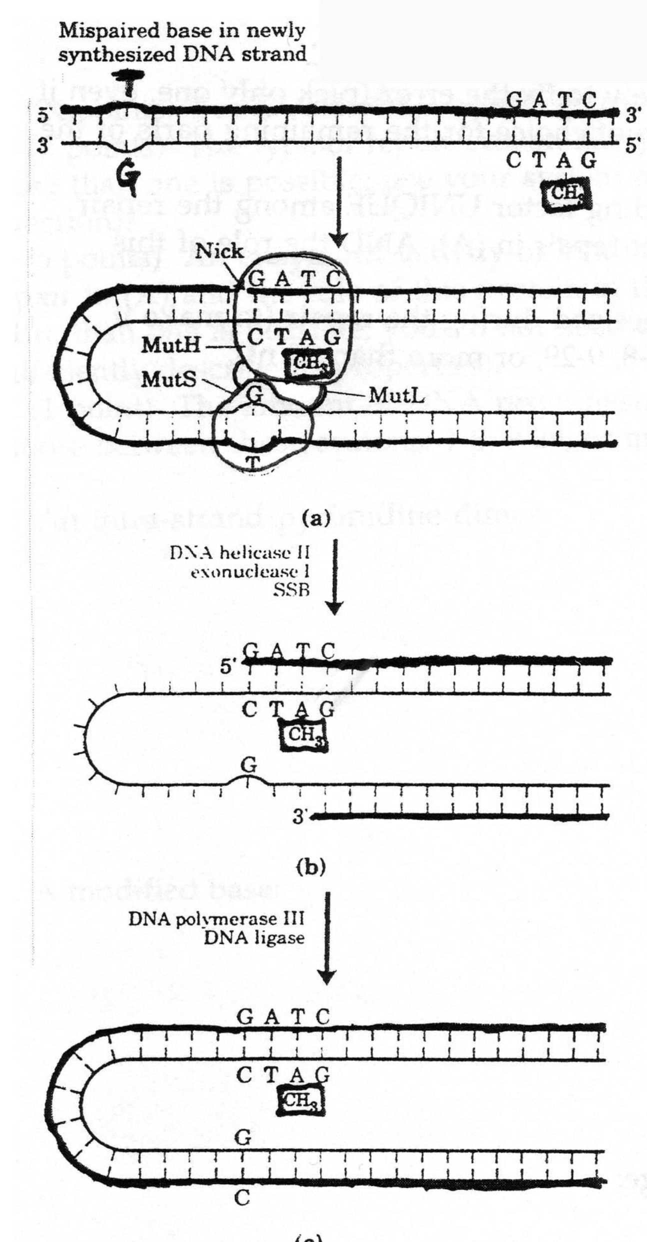
Mismatch repair

Which strand is new and which is the parent?





Mut S binds mismatch
Mut L links Mut S to Mut H
Mut H binds 7-meGATC, recognizes the parental strand,

Mismatch repair -- Recognition

Which strand is new and which is the parent? The mutation is in the new strand!

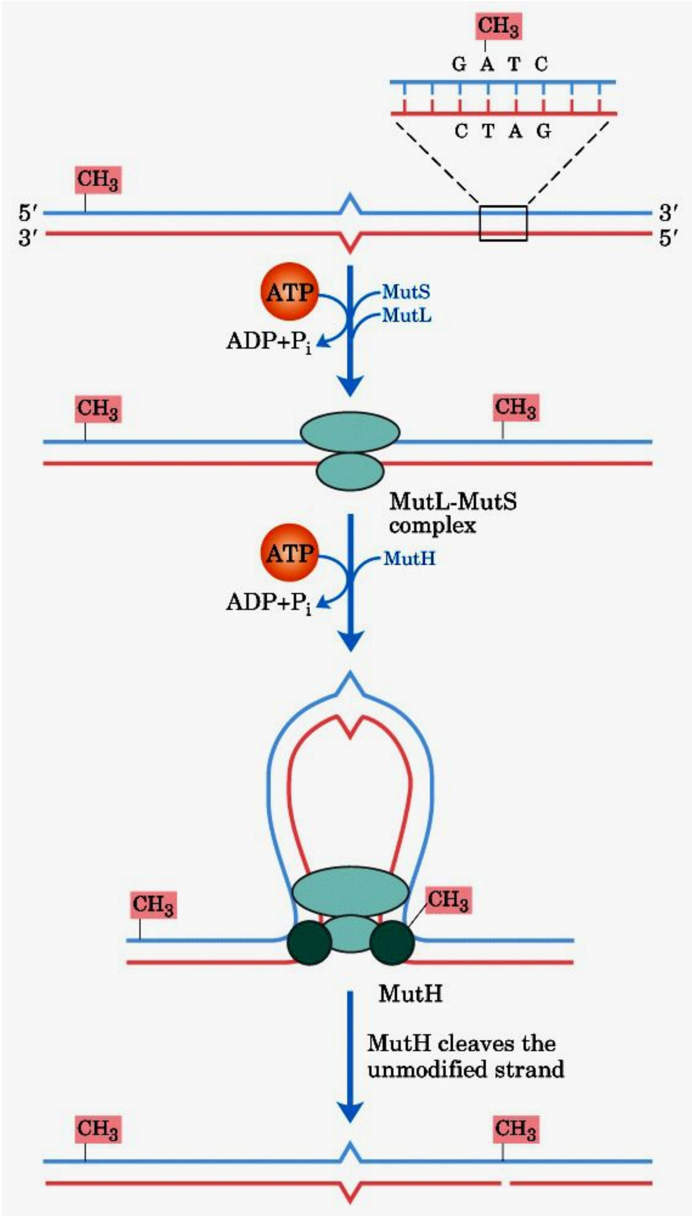
A-CH_3 marks the parental strand!

MutS - Binds mismatch

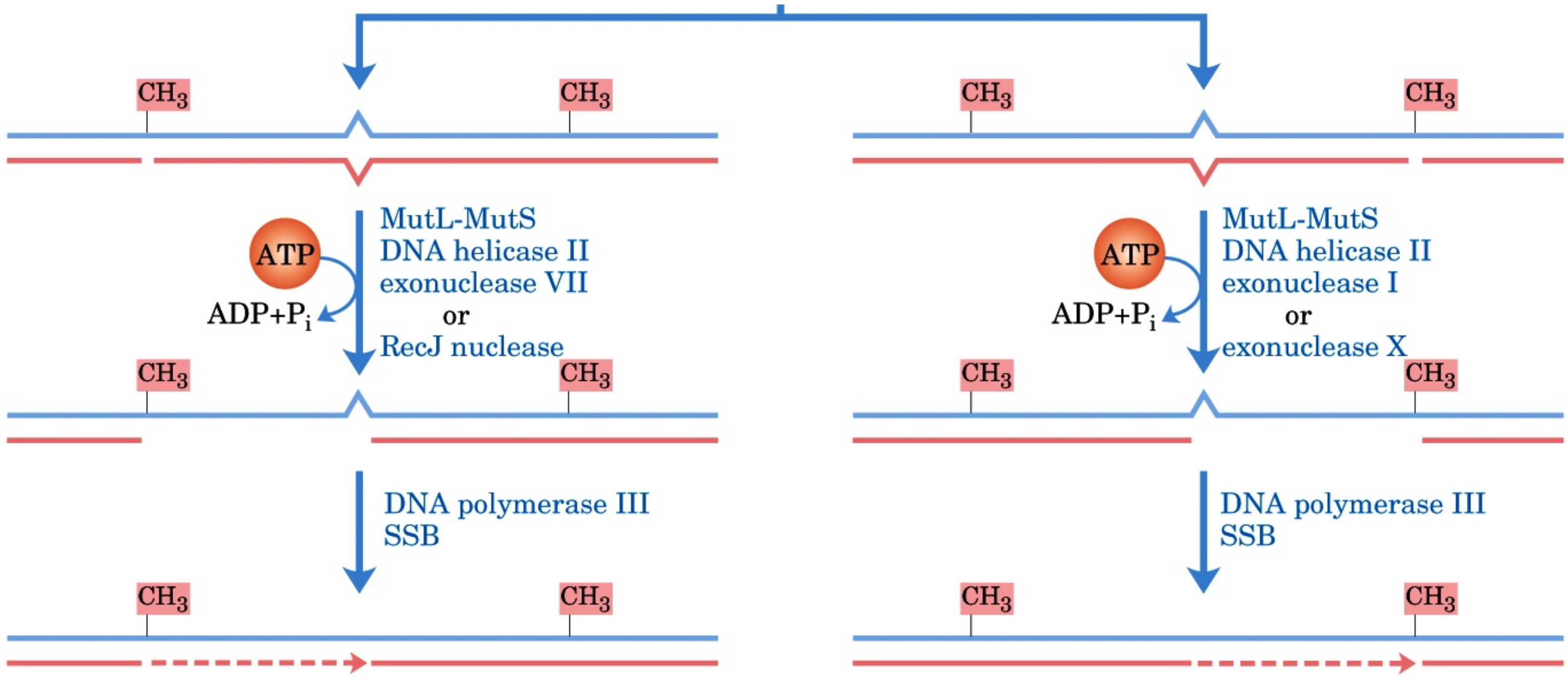
MutL - links MutH and MutS

MutH - Binds G^{me}ATC

DNA is threaded through the MutS/MutL complex. The complex moves simultaneously in both directions along the DNA until it encounters a MutH protein bound at a hemimethylated GATC sequence. MutH cleaves the unmethylated strand on the 5′ side of the G in the GATC sequence.

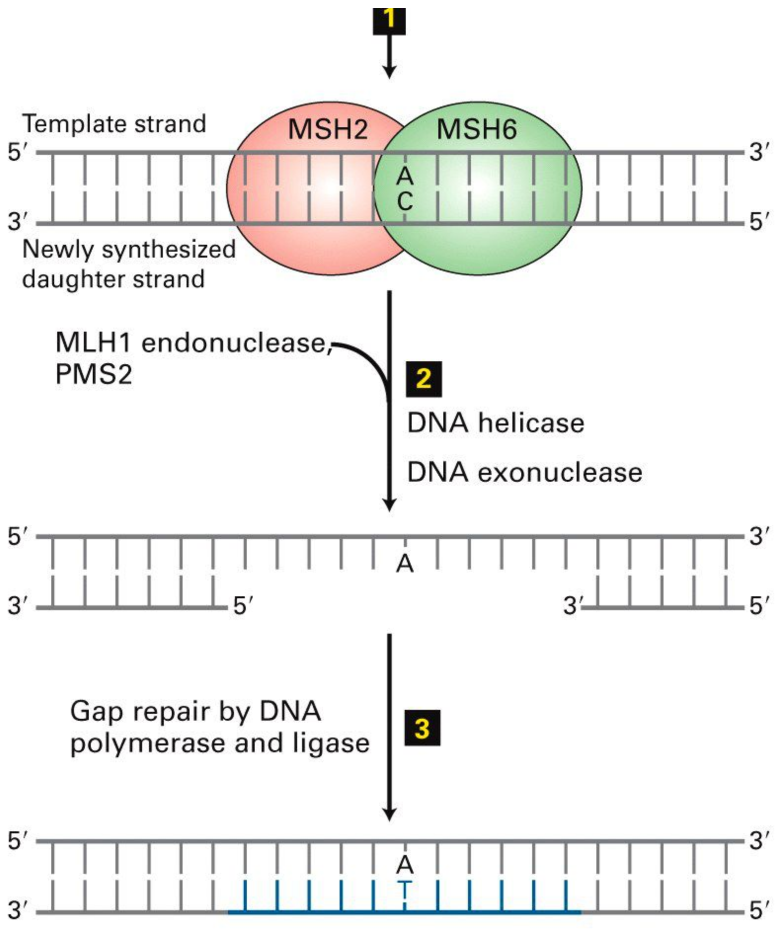


Mismatch repair -- Resolution



1. The combined action of DNA helicase II, SSB, and one of many different exonucleases (only two are labeled) removes a segment of the new strand between the MutH cleavage site and a point just beyond the mismatch.
2. The resulting gap is filled in by DNA polymerase III, and the nick is sealed by DNA ligase.

Mismatch repair -- MSH proteins -- eukaryotes



Defects in mismatch excision repair lead to colon and other cancers.

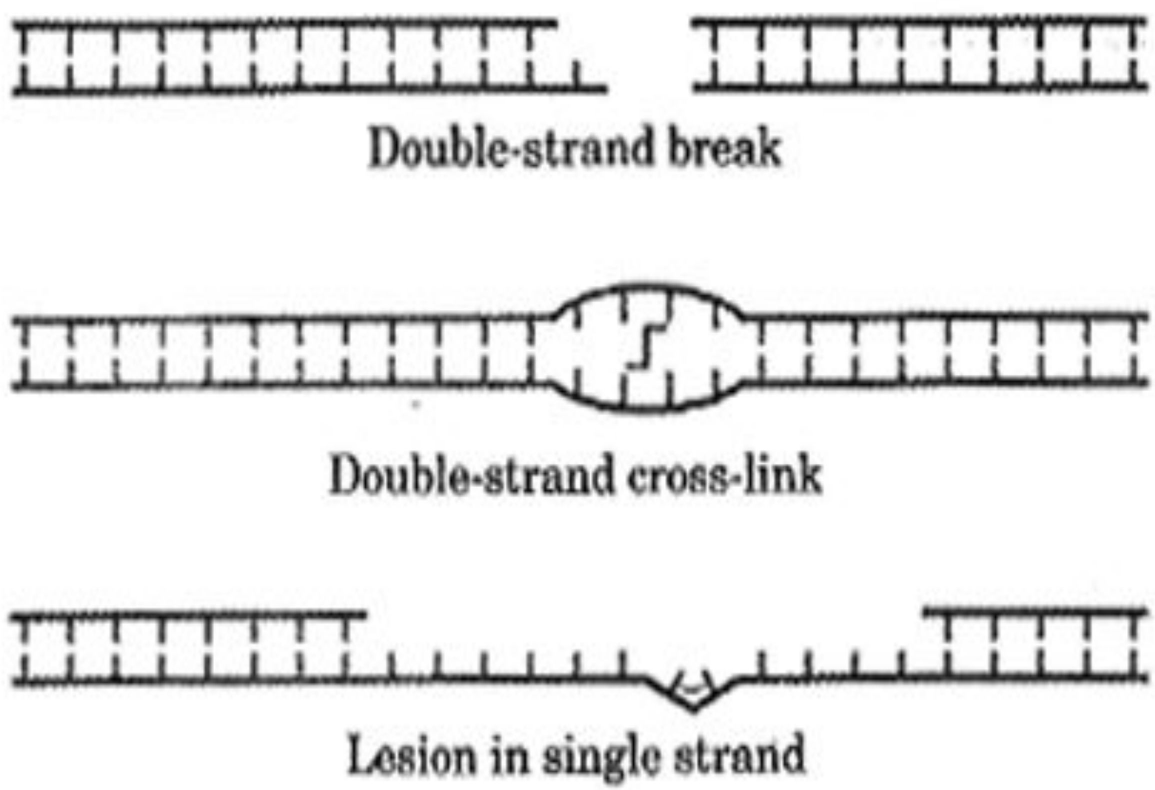
1. MSH2:MSH6 complex binds the mismatch and identifies newly synthesized strand.

2. MLHI endonuclease and other factors such as PMS2 bind, recruiting a helicase and exonuclease, which together remove several nucleotides including the lesion.

3. The gap is filled by Pol δ and sealed by DNA ligase.

Double-strand break repair

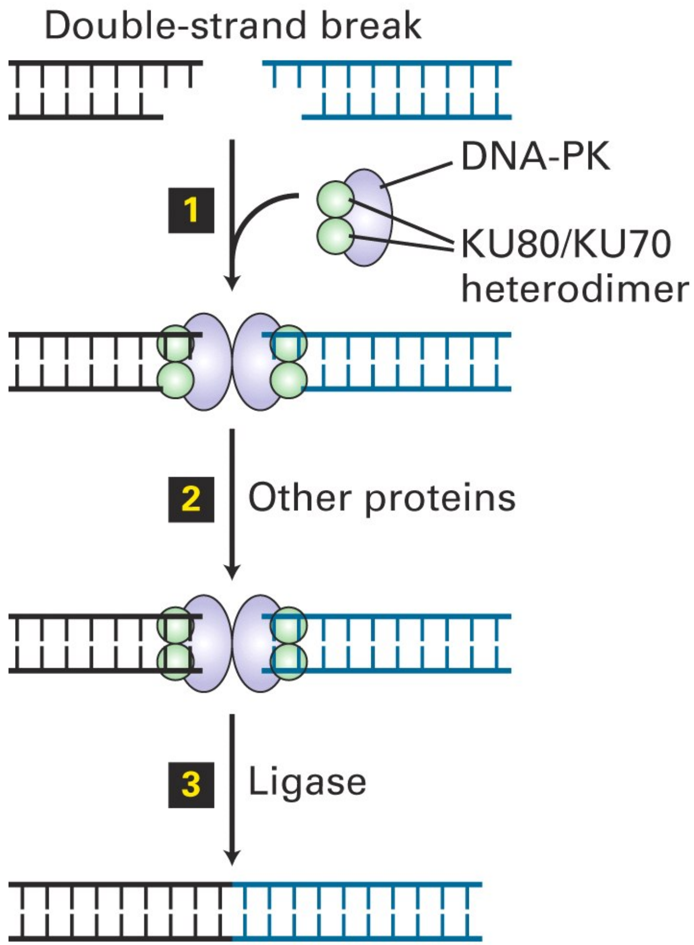
- Induced by ionizing radiation, transposons, topoisomerases, homing endonucleases, and mechanical stress on chromosomes



NO TEMPLATE FOR REPAIR!!

Double-strand break repair

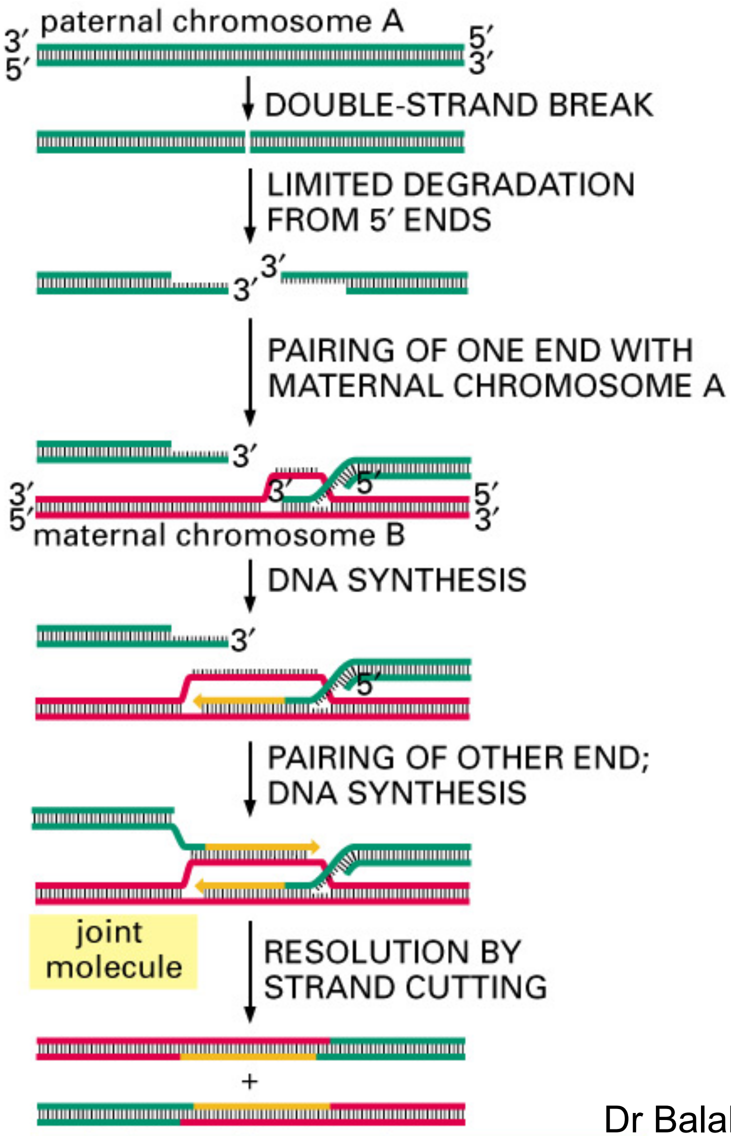
Two basic mechanisms: End-joining and Recombination

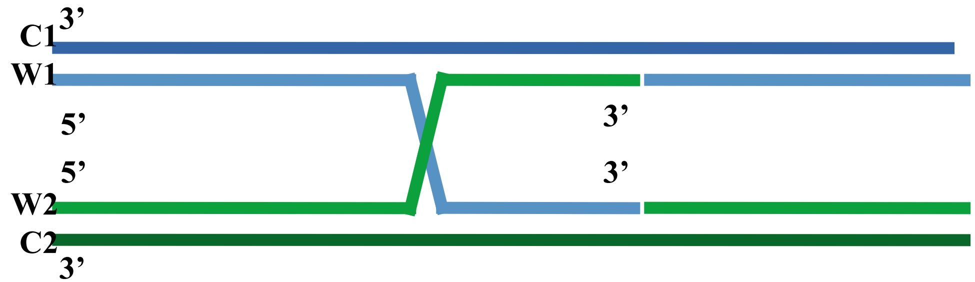


The end-joining pathway of ds break repair is mutagenic, because it removes several base pairs at the break site.

Mediated by Ku proteins.

Double-strand break repair -- Homologous recombination pathways





Strand exchange with nicks

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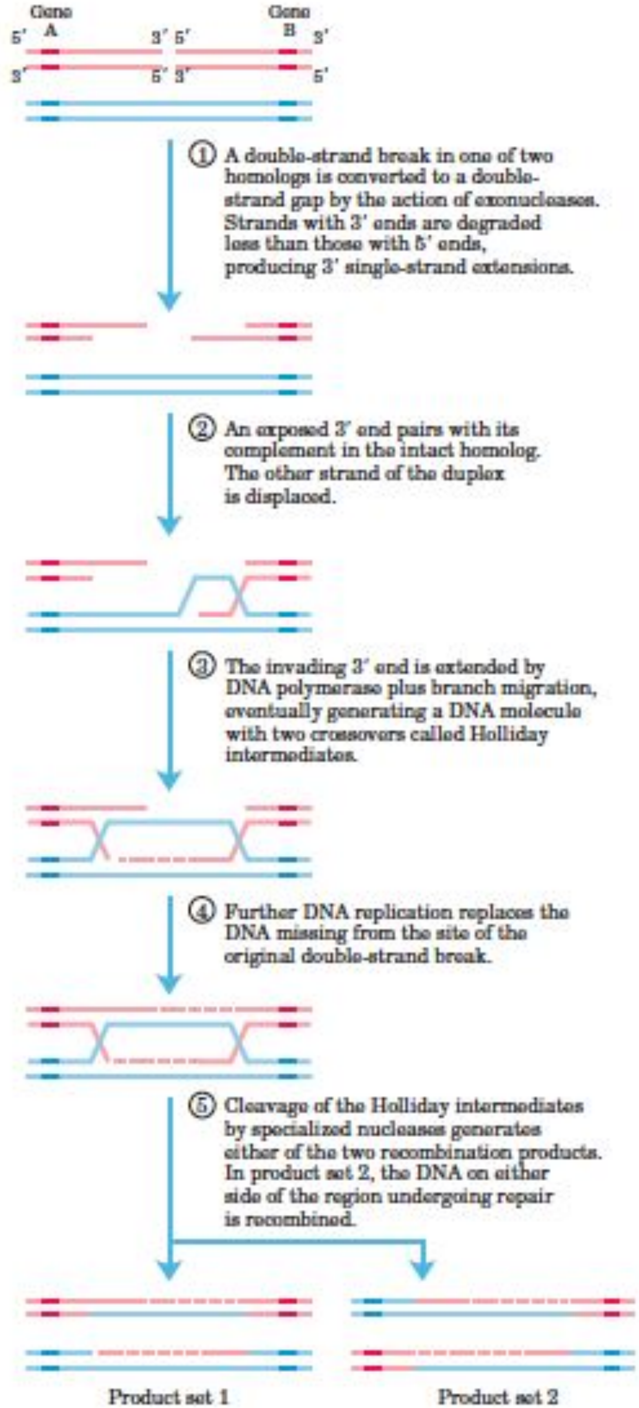


Gene A
Gene B
5′
3′
3′ 5′
5′ 3′
3′
5′
① A double-strand break in one of two homologs is converted to a double-strand gap by the action of exonucleases. Strands with 3′ ends are degraded less than those with 5′ ends, producing 3′ single-strand extensions.
② An exposed 3′ end pairs with its complement in the intact homolog. The other strand of the duplex is displaced.
③ The invading 3′ end is extended by DNA polymerase plus branch migration, eventually generating a DNA molecule with two crossovers called Holliday intermediates.
④ Further DNA replication replaces the DNA missing from the site of the original double-strand break.
⑤ Cleavage of the Holliday intermediates by specialized nucleases generates either of the two recombination products. In product set 2, the DNA on either side of the region undergoing repair is recombined.
Product set 1
Product set 2

DNA repair defects cause disease

TABLE 23-1 Some Human Hereditary Diseases and Cancers Associated with DNA-Repair Defects

Disease	DNA-Repair System Affected	Sensitivity	Cancer Susceptibility	Symptoms
PREVENTION OF POINT MUTATIONS, INSERTIONS, AND DELETIONS				
Hereditary nonpolyposis colorectal cancer	DNA mismatch repair	UV irradiation, chemical mutagens	Colon, ovary	Early development of tumors
Xeroderma pigmentosum	Nucleotide excision repair	UV irradiation, point mutations	Skin carcinomas, melanomas	Skin and eye photosensitivity, keratoses
REPAIR OF DOUBLE-STRAND BREAKS				
Bloom's syndrome	Repair of double-strand breaks by homologous recombination	Mild alkylating agents	Carcinomas, leukemias, lymphomas	Photosensitivity, facial telangiectases, chromosome alterations
Fanconi anemia	Repair of double-strand breaks by homologous recombination	DNA cross-linking agents, reactive oxidant chemicals	Acute myeloid leukemia, squamous-cell carcinomas	Developmental abnormalities including infertility and deformities of the skeleton; anemia
Hereditary breast cancer, BRCA-1 and BRCA-2 deficiency	Repair of double-strand breaks by homologous recombination		Breast and ovarian cancer	Breast and ovarian cancer

SOURCES: Modified from A. Kornberg and T. Baker, 1992, *DNA Replication,* 2d ed., W. H. Freeman and Company, p. 788; J. Hoeijmakers, 2001, *Nature* **411**:366; and L. Thompson and D. Schild, 2002, *Mutation Res.* **509**:49.

Mutations

Mutation:

- A permanent transmissable change in the genetic material, usually in a single gene

Types of Mutations:

- Genomic: abnormal chromosome number (monosomy, polysomy, aneuploidy)
- Chromosomal: abnormal chromosome structure
- Gene: DNA sequence changes in specific genes

Genomic Mutation:
Variation In Chromosome Number

Euploidy

- Normal variations of the number of complete sets of chromosomes
- Haploid, Diploid, Triploid, Tetraploid, etc…

Aneuploidy

- Variation in the number of particular chromosomes within a set
- Monosomy, trisomy, polysomy

Numerical Aberration

- **Autosomal**

- Trisomies: 1 chromosome extra (e.g. trisomy 21-13-18)

- Monosomies: 1 chromosome is missing

- **Sex chromosome**

- Klinefilter syndrome (47, XXY male)

- Turner syndrome (45, XO female)

Numerical abnormalities

Trisomy i.e. 47 chromosomes

- Trisomy 21 (the extrachromosome is No 21)
- Klinefelter syndrome (47, XXY male)

Monosomy i.e. 45 chromosomes

- Monosomy 21
- Turner syndrome (the missing chromosome is X in female : 45, X or 45 XO)

TABLE 8.1

Aneuploid Conditions in Humans

Condition	Frequency	Syndrome	Characteristics
Autosomal			
Trisomy 21	1/800	Down	Mental retardation, abnormal pattern of palm creases, slanted eyes, flattened face, short stature
Trisomy 18	1/6,000	Edward	Mental and physical retardation, facial abnormalities, extreme muscle tone, early death
Trisomy 13	1/15,000	Patau	Mental and physical retardation, wide variety of defects in organs, large triangular nose, early death
Sex Chromosomal			
XXY	1/1,000 (males)	Klinefelter	Sexual immaturity (no sperm), breast swelling
XYY	1/1,000 (males)	Jacobs	Tall
XXX	1/1,500 (females)	Triple X	Tall and thin, menstrual irregularity
X0	1/5,000 (females)	Turner	Short stature, webbed neck, sexually undeveloped

Chromosomal Mutation:
Structural abnormalities

1) **Translocation** :

the transfer of a chromosome or a segment of it to a non-homologous chromosome.

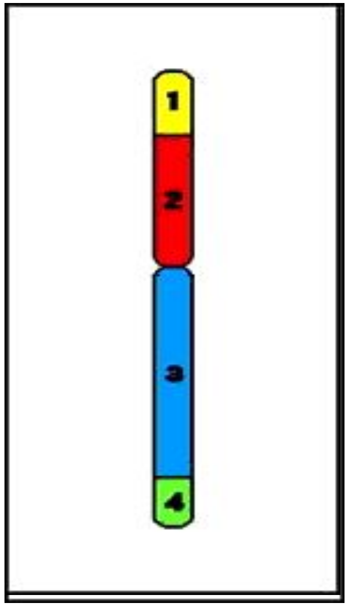
2) **Deletion** : loss of a portion of a chromosome.

3) **Ring chromosome**

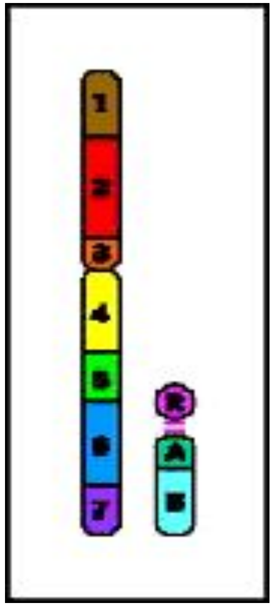
4) **Duplication** : extra piece of a chromosome.

5) **Inversion** : fragmentation of a chromosome followed by reconstitution with *a* section inverted.

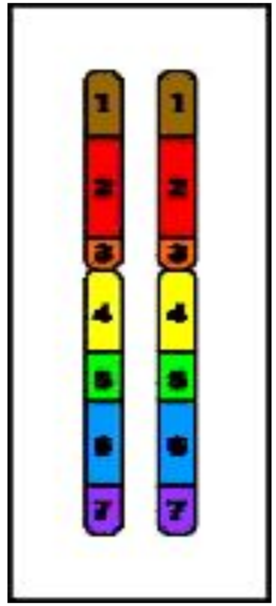
6) **Isochromosomes**: division of chromosome at centromere transversely instead of longitudinally



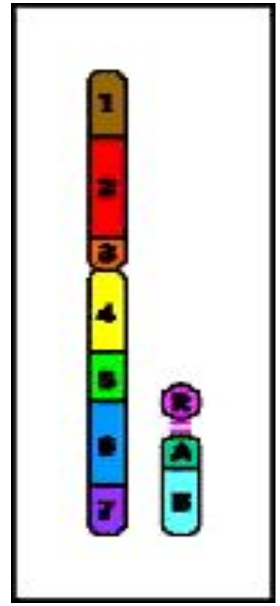
Ring chromosome



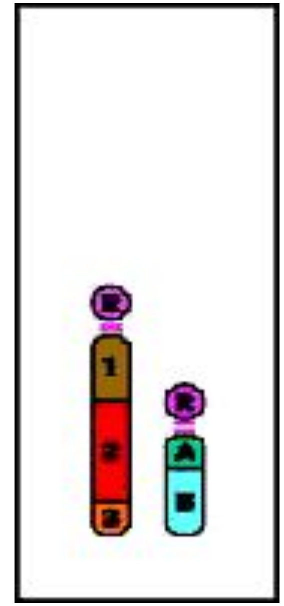
Insertion



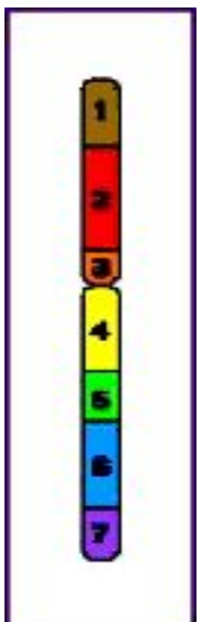
Duplication



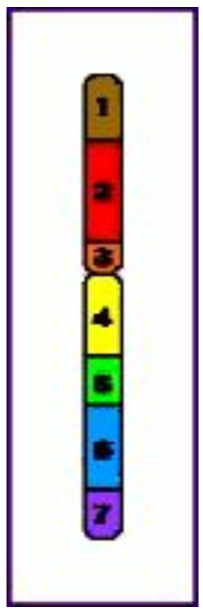
Reciprocal translocation



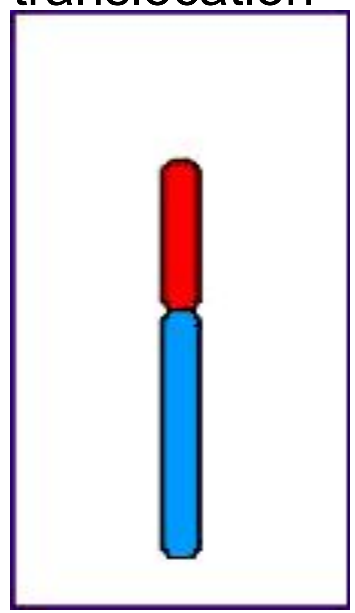
Robertsonian translocation



Paracentric inversion



Pericentric inversion



Isochromosme

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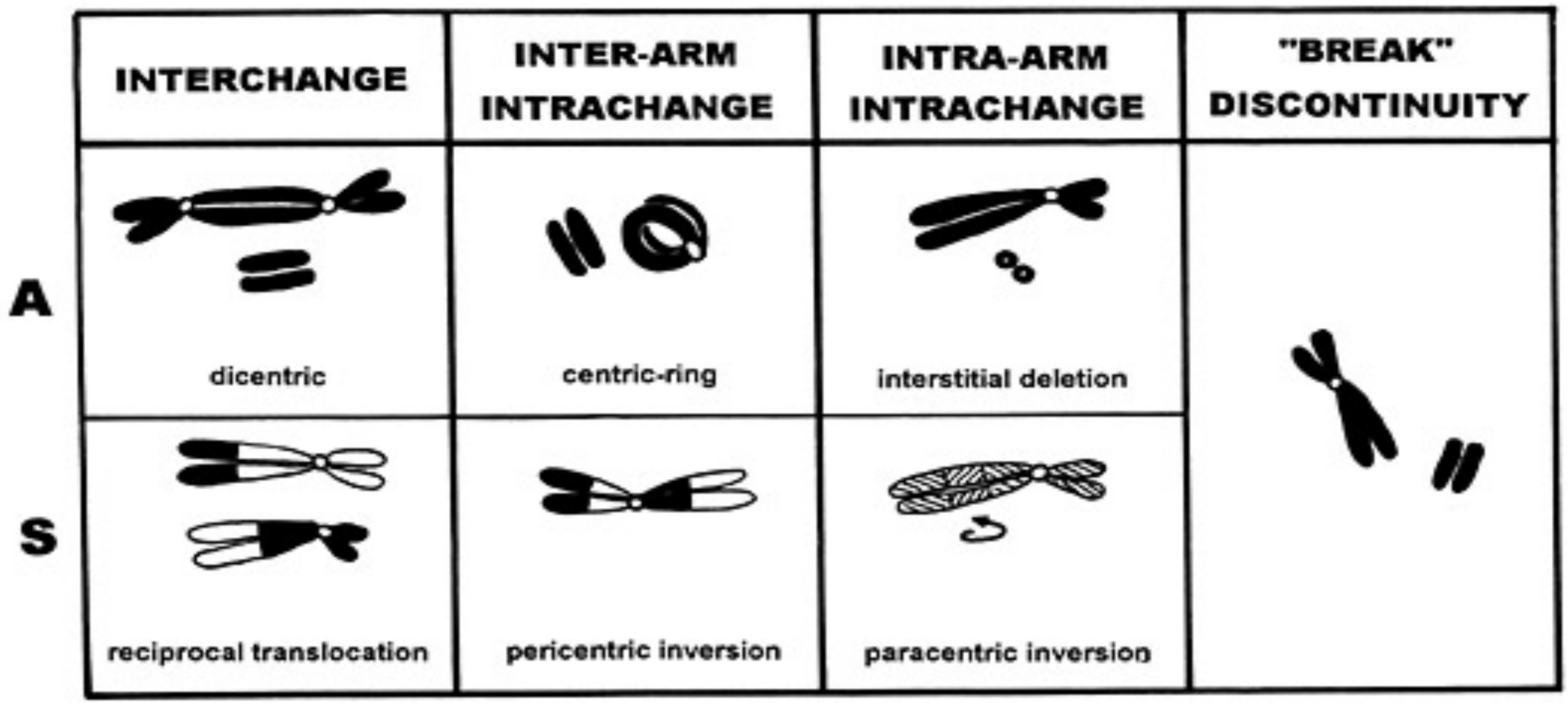
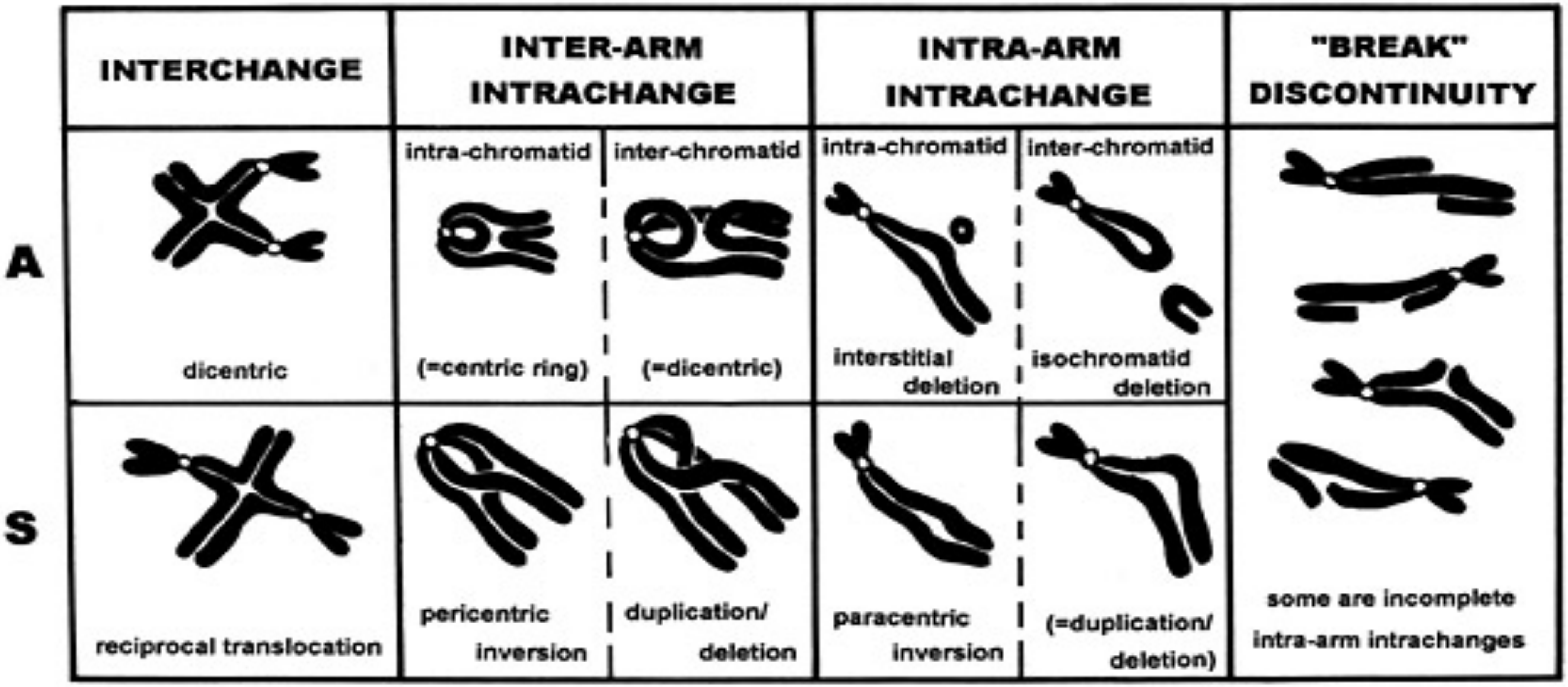
Examples of 2-lesion *Chromosome-type* aberrations

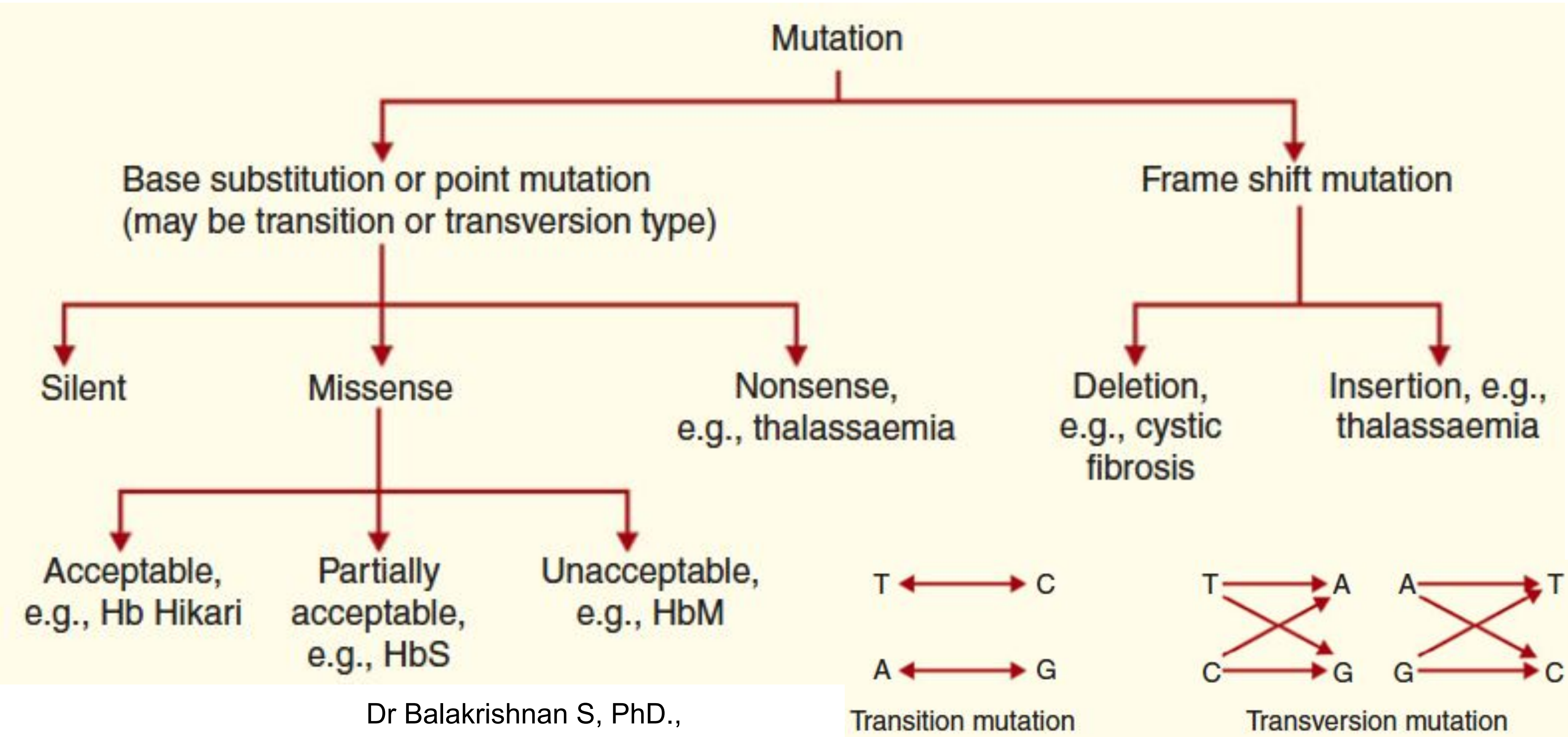


Examples of 2-lesion *Chromatid-type* aberrations



Gene Mutations

- Gene Mutations - or changes to the nucleotide sequence of DNA - can arise in a number of ways.
- **Types of Gene Mutations:**



Gene Mutations

Based on severity, the mutation can further divided in to the followings:

- **Silent mutation**
- **Non sense mutation**
- **Missense mutation**

Read more…………

……………………………Thank you